

Robert Casey Lafferty

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Gameplay programmer with professional experience in AAA and Indie game studios using C++ and a focus in game AI.

Game Credits

AAA: Call of Duty: Black Ops 6 (Proprietary Engine)	Oct 2024
AAA: Call of Duty: Modern Warfare III (Proprietary Engine)	Nov 2023
Indie: Social Sidekick (Unity Engine)	March 2023

Computer Skills

Languages	C++, C#, Java, C, Python, Lua, MySQL, SQL Server
Development Tools	Unreal Engine, Unity Engine, Git, Perforce, PlasticSVN, Audacity, Mixamo
Game AI Skills	UE State Trees, UE Behavior Trees, UE EQS, UE Perception Systems, State Machines

Professional Work Experience

Game Programming Instructor (Unreal Engine 5 C++, Unity C#, LOVE Engine Lua) Academy of Art University, Remote	Aug 2024 – Present
- Created custom assignments to apply lesson concepts to real-world game development scenarios - Developed long-term solutions to improve student learning retention for all course concepts	
Gameplay Engineer – AI (Unreal Engine 5 C++/Blueprint) Deviant Games (Contract), Remote	March 2024 – Jan 2025
- Prototyped new AI mechanics, behaviors, and components to enhance in-game mechanics and player experiences - Designed technical subsystems to coordinate AI behavior intended to drive specific player experiences	
Gameplay Engineer – Call of Duty (C++, Python, Proprietary Tech) Treyarch, Playa Vista, CA	Oct 2022 – Sept 2023
- Engineered 22 gameplay mechanics to empower players' experience in the proprietary Call of Duty engine - Optimized workflow for my team and collaborators by centralizing information about a proprietary tool	
Software Engineer II – Tools & UI (C#, C++, MS SQL Server) AGS, Duluth, GA	Dec 2021 – July 2022
- Designed a brand new Windows service to assist in legacy support for older servers - Enhanced QA's workflow by improving the UI and back-end functionality for existing software suites	
Software Engineer – UI (C++) Naval Surface Warfare Center, Dahlgren, VA	March 2021 – Sept 2021
- Implemented UI and back-end fixes to mission-critical C++ code for government weapons systems - Streamlined new hire onboarding documentation and exercises to improve future new hire experiences	

Game Dev Projects

Unreal Engine Sandbox (Unreal Engine 5 C++/Blueprint) – Side Project (Link)	Sept 2023 – Present
3D third-person sandbox project used to explore game ideas, AI, player controls, VFX, animation, and cinematics - Designed enemy AI behaviors including patrol, detection, and pursuit using Unreal Engine perception systems	
Wrath of Raoden (Unity C#) – Side Project (Link)	Dec 2020 – Feb 2024
2D roguelike dungeon crawler where the player must conquer randomized dungeons to save the world - Engineered mobile game from scratch with a focus in gameplay experience, AI, and player mechanics	

Education

University of Nebraska-Lincoln , Lincoln, NE Master of Science in Computer Science CS GPA: 3.39	Dec 2020
Kansas State University , Manhattan, KS Bachelor of Science in Computer Science Minor: Music CS GPA: 3.26	May 2017

Honors & Activities

Co-Founder, President – UNL Game Development Club	Aug 2018 – Dec 2019
Eagle Scout	Sept 2012